SMART DISPLAY MODULE SPECIFICATION

1.5 Inch Smart Knob Display with Touch	
Model:	UEDX46460015-WB-A
Version:	V1.1
Date:	2024-10-16

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Customer Confirmation

Approved by	Notes

REVISION HISTORY

Revision	Date	Contents of Revision Change	Remark
V1.0	20241010	Preliminary release	
V1.1	20241016	Add GitHub project links, and environment confi links	

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1. Introduction

1.1 Features

Brief Info:

1) Outline Dimension: ϕ 51.6 Round
2) Interaction Method: Rotate and Press
3) Shell Color: Black/Silver/Customized
4) Power: DC 5V, 100mA

System

1) OS: RTOS
2) CPU: ESP32-S3 240Mhz
3) RAM: 8MB
4) Flash: 16MB
5) Interface: UART/USB
6) Support 2.4GHz Wi-Fi、BLE 5、BLE Mesh

Note: The Bluetooth and WIFI functions are still under test, we will complete them as soon as possible, please forgive the inconvenience caused to you

Display

1) Size: 1.5 Inch
2) Resolution: 466*466
3) Mode: AMOLED
4) Driver IC: CO5300AF-42
5) Interface type: QSPI
6) Brightness: 1000 cd/m^2
7) Touch IC: CST820

Other

1) Operation Temperature：-20~60℃
2) Storage Temperature：-30~70℃

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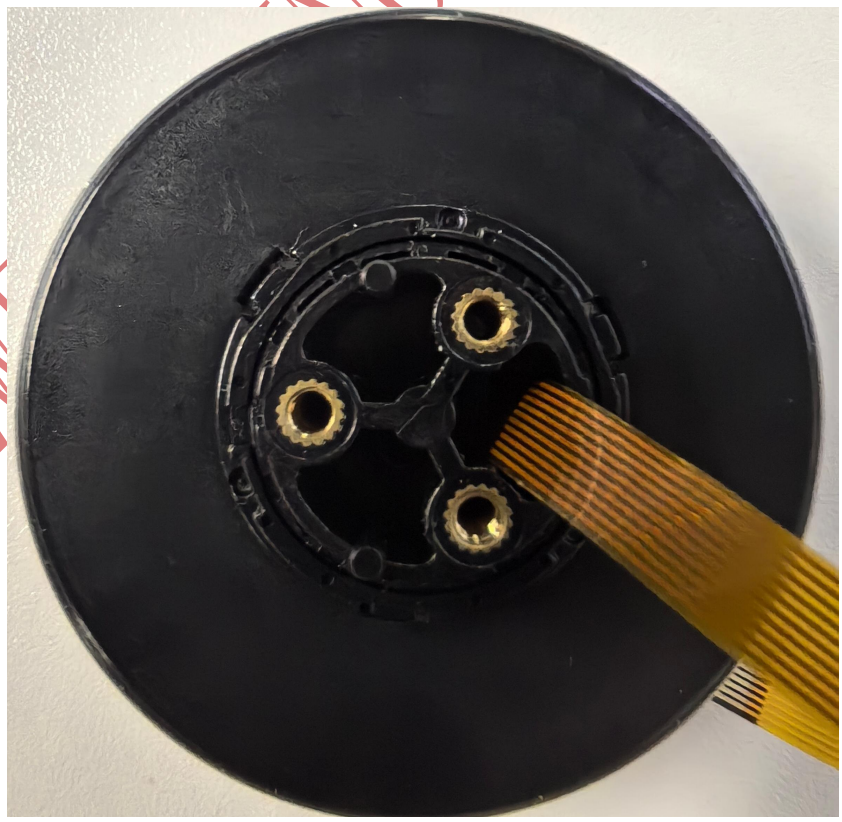
1.2 Appearance picture



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2. Product information

2.1 Connector Interface Description

Pin NO.	Symbol	Description	Voltage Range	Remarks
1	VCC	Power 5V	5V	
2	GPIO38	-	-	
3	GND	Grounds	0V	
4	GPIO40	GPIO40	-	
5	GPIO39	GPIO39	-	
6	RX	UART Receive	0-3.3V	
7	TX	UART Transmit	0-3.3V	
8	CHIP-EN	CHIP-EN	0-3.3V	
9	D+	USB D+	3.3V	
10	D-	USB D-	3.3V	

The connector specifications is 10PIN 0.5mm pitch

2.2Display interface description

Pin No.	Symbol	I/O	Description
1	**VCI_EN**	P	Power supply
2	**NC**	-	-
3	**NC**	-	-
4	**NC**	-	
5	**QSPI_SI03**	I	QSPI Data3 input pin
6	**QSPI_SI02**	I	QSPI Data2 input pin
6	**QSPI_SI01**	I	QSPI Data1 input pin
7	**QSPI_SI00**	I/O	QSPI Data0 input pin
9	**NC**	-	-
10	**SCL**	I	SPI pclk input signal
11	**NC**	-	-

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12	**CS**	I	Chip selection pin. Low-active
13	**GND**	P	Power Ground
14	**TE**	O	Tearing effect signal is used to synchronize MCU to frame memory
15	**IOVCC**	P	Power supply to the internal logic power regulator
16	**MTP(NC）**	-	-
17	**VBAT**	P	Power supply
18	**RST**	I	The signal will reset the LCM, Signal is active low.
19	**TP_SCL**	I	I2C clock signals for CTP
20	**TP_SDA**	I/O	I2C data signal for CTP
21	**TP_RST**	I	The signal will reset the CTP,Signal is active low
22	**TP_INT**	I	Interrupt signals for CTP
23	**TP_VCC**	P	Power supply for TP
24	**GND**	P	Power Ground

2.3 Display Information

Item	Parameter	Description
Color	16.7M colors	24bits
AA	38.3052*38.3052	1.51 inch
Resolution	466*466	Round
Brightness	1000cd/m2	-
Interface Mode	QSPI	-
Driver IC	CO5300AF-42	-
Pixel Driving element	AMOLED	-

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2.4 Voltage & Current

Item	Conditions	Min	Typ	Max	Unit
Power Voltage	DC	4. 0	5.0	5.5	V
Operation Current	VCC= +5V, Maximum backlight current	50	100	150	mA
	VCC= +5V,backlight off	-	50	-	mA
Recommended power supply:5V 1A DC					

2.5 Reliability Test

Item	Conditions	Min	Typ	Max	Unit
Working Temperature	60%RH at 5V voltage	-20	25	60	C
Storage Temperature	---	-30	25	70	C
Working Humidity	25°C	10%	60%	90%	RH
ESD	---	Contact: ±4KV Air: ±8KV			KV

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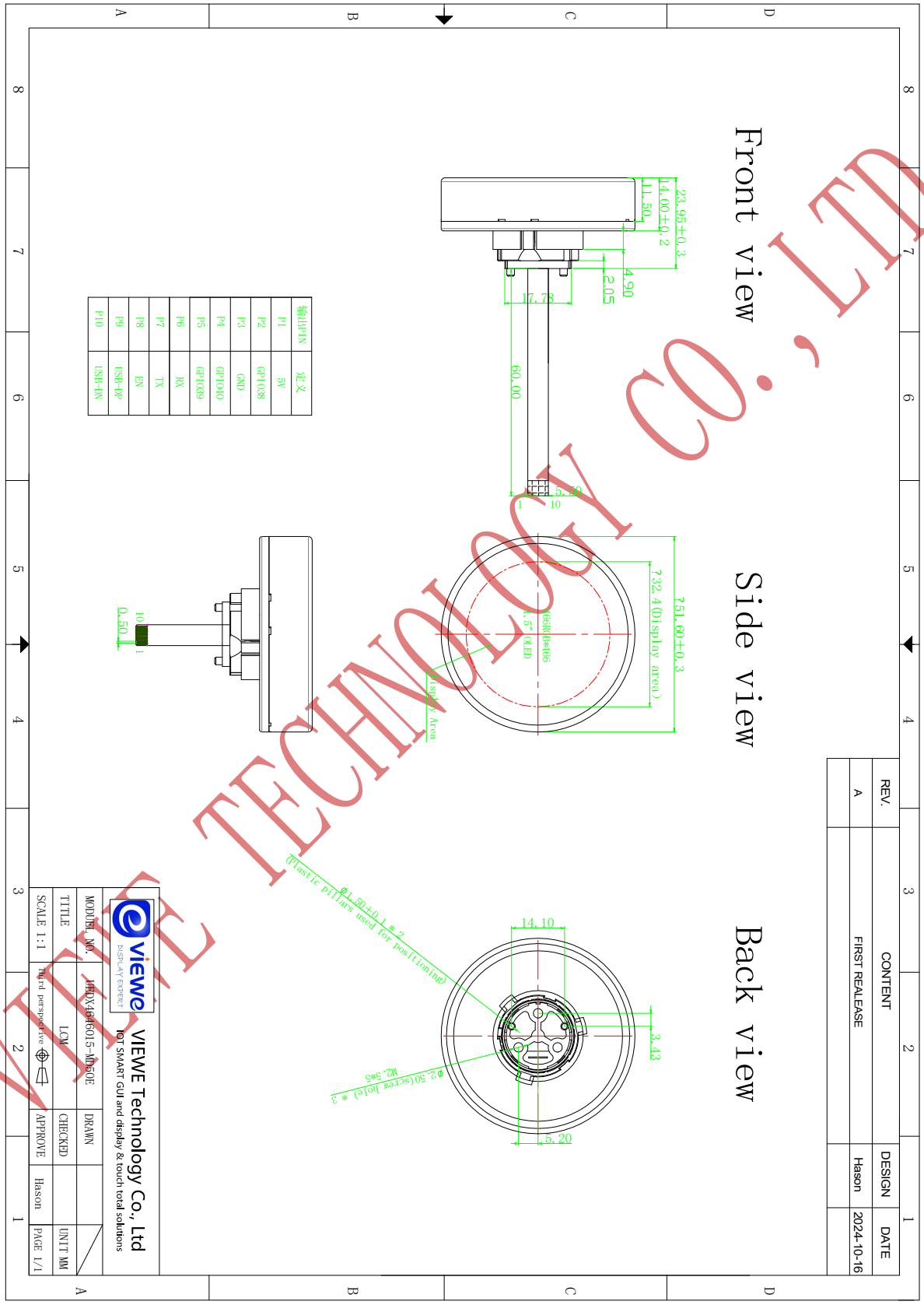
3. MECHANICAL DRAWING



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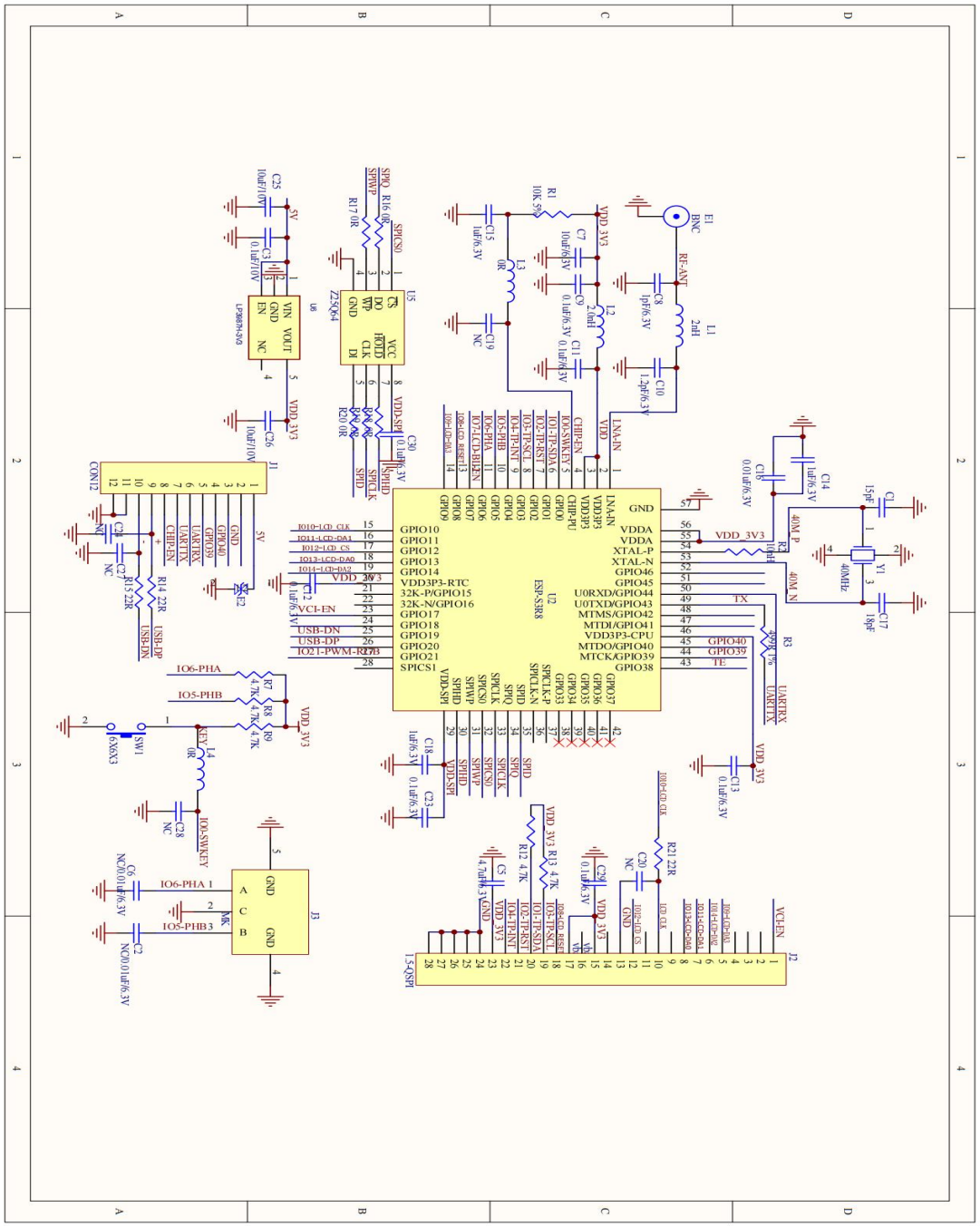
4. Schematic


U2
ESP-S3R8
U5
Z25Q64
U6
LP38854-SD-3
J1
CON12
J2
1.5-QSPI
J3
E1
BNC
Y1
40MHz
SW1
0.6X3.5
RF-ANT
40M_P
40M_N
VDD_3V3
VCI-EN
USB-DP
USB-DN
IO6-PHA
IO5-PHB
IO6-PHA 1
IO5-PHB3
IO5-SWKEY
TX
GPIO40
GPIO39
TE
UARTRX
UARTTX
CHIP-EN
GPIO39
GPIO40
GND
5V
SPICS0
SPIWP
SPIHD
SPICLK
SPID
SPIQ
VDD-SPI
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
VDD3P3-RTC
32K-P/GPIO15
32K-N/GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
SPICS1
GND
VDDA
VDDA
XTAL-P
XTAL-N
GPIO46
GPIO45
U0RXD/GPIO44
U0TXD/GPIO43
MTMS/GPIO42
MTDI/GPIO41
VDD3P3-CPU
MTDO/GPIO40
MTCK/GPIO39
GPIO38
LNA-IN
VDD3P3
VDD3P3
CHIP-PU
GPIO0
IO0-SWKEY
IO1-TP-SDA
IO2-TP-RST
IO3-TP-SCL
IO4-TP-INT
IO5-PHB
IO6-PHA
IO7-LCD-BHEN
IO8-LCD_RESET
IO9-LCD-RS
GPIO9
GPIO8
GPIO7
GPIO6
GPIO5
GPIO4
GPIO3
GPIO2
GPIO1
GPIO37
GPIO36
GPIO35
GPIO34
GPIO33
SPICLK-P
SPICLK-N
SPID
SPIQ
SPICLK
SPICS0
SPIWP
SPIHD
VDD-SPI
IO10-LCD_CLK
IO11-LCD-DA1
IO12-LCD_CS
IO13-LCD-DA0
IO14-LCD-DA2
IO21-PWM-RGB
VDD_3V3
VDD_3V3
VDD_3V3
VDD_3V3
VDD_3V3
VDD_3V3
5V
5V
GND
GND
GND

5. Related downloads

5.1 Arduino and IDF relevant information

https://github.com/VIEWESMART/UEDX46460015-MD50ESP32-1.5inch-Touch-Knob-Display

5.2 Libraries required for Arduino

https://github.com/VIEWESMART/UEDX46460015-MD50ESP32-1.5inch-Touch-Knob-Display/tree/main/Libraries

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